

ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SEMESTER PROJECT FALL 2025 (8 ECTS)
MASTER IN COMPUTATIONAL SCIENCE AND ENGINEERING

ENVIRONMENTAL COMPUTATIONAL SCIENCE AND EARTH OBSERVATION
LABORATORY (ECEO)

Topography-aware Temperature Forecasting in Europe

Author:
Jan ZGRAGGEN

Daily Supervisor:
Chang XU

Supervisor:
Prof. Devis TUIA

EPFL

Contents

1	Introduction	3
2	Related work	4
2.1	Weather forecasting	4
2.2	Geography-informed learning	4
2.2.1)	Research Gap	5
3	Methods	5
3.1	Definitions	5
3.1.1)	Grid Variables	5
3.1.2)	Grid Variable Kernels	6
3.1.3)	Task description	6
3.2	Baseline	7
3.3	Simple Fusion	7
3.4	(geographical) Implicit Neural Representation - gINR	8
3.4.1)	GeoFAR-style gINR	8
3.5	Explicit Empirical Consistency Regularization (EECR)	9
3.5.1)	Histogram-based EECR	10
3.5.2)	Space-Dependent EECR	12
3.5.3)	Geography-Aware Space-Dependent EECR	12
4	Experiments	12
4.1	Experimental Setup	12
4.1.1)	Data	12
4.1.2)	Implementation Details	13
4.2	Results	13
4.2.1)	Main Comparison	13
4.2.2)	Bias Analysis	14
4.3	Discussion	15
4.3.1)	General Observations	15
4.3.2)	Model Comparisons	15
5	Conclusion	16
5.1	Summary	16
5.2	Outlook	17
5.2.1)	Geography-informed Learning	17
5.2.2)	Extension of the EECR Framework	17
5.3	Limitations	18
6	Code availability	18
7	Acknowledgments	18
A	Appendix	23
A.1	Model naming conventions	23
A.2	Regularization naming conventions	23
A.3	Validation metric plot for reported experiments	23

List of Figures

1	Visualization of bias in alpine regions	3
2	Visualization of the <i>History</i> , <i>Window</i> , and <i>Lead-time</i> parameters in the forecasting task	7
3	Pipeline of the gINR framework for geographical positional encoding with sinusoidal-harmonics terrain encoding and SIREN encoder.	8
4	Histogram plots of $ \Delta T $ in $ \Delta H $ -bins and curve fitting.	11
5	$MAB(i, j)$ for selected models and their comparisons. Subfigures (e)–(g) show the relative improvement of models (a)–(d) in sequential comparisons.	14
6	Comparison of MSE and MSE + EECR ($\lambda = 0.1$) objectives.	16
7	RMSE validation metric plot of the methods reported in Table 2.	23

List of Tables

1	Model configuration parameters. $\mathcal{I} = 3h$ denotes the interval between measurements in the time series.	7
2	Test and validation Root Mean Square Error (RMSE) and Mean Absolute Bias (MAB) for the evaluated models. Lower values indicate better performance. The table is ordered by decreasing RMSE (indicated by the symbol $\downarrow$). Model and EECR abbreviations follow the naming conventions defined in Table 3 and Table 4 respectively, found in Appendix A.1 and A.2.	14
3	Overview of model naming conventions used throughout this work, with references to the corresponding method descriptions.	23
4	Overview of EECR naming conventions used throughout this work, with references to the corresponding method descriptions.	23

1 Introduction

The increasing frequency of occurrence of extreme weather events due to climate change [1, 2, 3] has heightened the importance of accurate weather forecasting [4, 5]. Machine learning has therefore been widely explored for weather forecasting in order to overcome the limitations of traditional modeling strategies such as general circulation models (GCMs) and numerical weather prediction (NWP) models. These limitations include high computational cost, low inference speed, and biases introduced by numerical discretization and parameterization schemes [6].

The rapidly growing volume of observational data available in recent years [7, 8, 9, 10] further motivates the adoption of machine learning to weather forecasting [11, 12]. Data-driven approaches have demonstrated performance comparable to that of numerical methods in many forecasting tasks, while providing higher efficiency and faster inference.

Most current data-driven models rely exclusively on observational data and do not explicitly account for physical relationships [4, 13, 14], which can lead to systematic biases in complex terrains. Atmospheric properties such as surface temperature are governed by physical processes driven by atmospheric dynamics [15], as well as by physical constraints such as terrain [16, 17]. These constraints are especially pronounced in complex regions such as the Alps, where measurements are strongly modulated by orographic effects. As a consequence, spatially varying biases have been observed in the forecasting of atmospheric variables [18]. Several studies indicate a correlation between forecast biases and topographic features [19, 20].

It is therefore important to explicitly account for physical constraints in order to improve the physical fidelity of forecasting results and to compensate for potential missing information with respect to the true data distribution. Incorporating physical priors into data-driven models has thus become an important research objective [21]. One domain in which physical constraints—particularly orographic information—are crucial is (near-)surface temperature forecasting. Even small spatial variations can correspond to large altitude differences, resulting in substantially different climate conditions and surface temperatures. Furthermore, accurate surface temperature prediction is highly relevant for societal applications and for understanding phenomena with direct correlations, such as glacier melting and snow-albedo effects, as well as more complex interactions with precipitation, vegetation, or fauna [22, 23, 24].

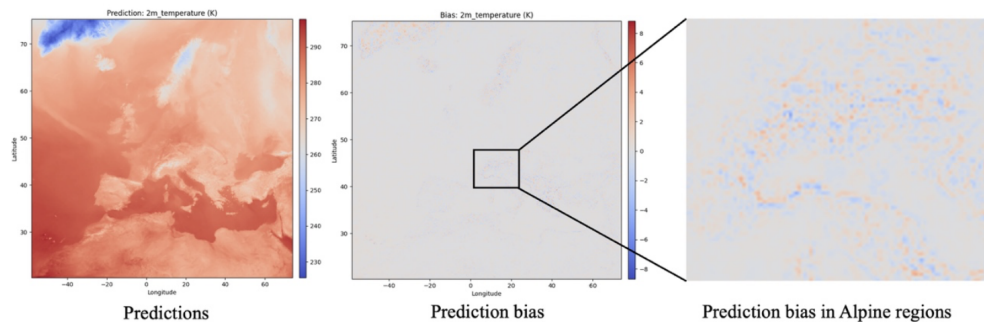


Figure 1: Visualization of bias in alpine regions

While strategies for incorporating physical constraints into ML models have been described in the literature, few studies have examined their application in terrain-sensitive prediction domains.

In this project, we therefore explore three approaches to improve the integration of orographic information: simple fusion, location embeddings, and loss regularization. We evaluate these methods on temperature measurements over Europe using the CERRA dataset [25]. For the data-driven baseline model ViT (subsection 3.2), see Figure 1 [26], we observe a severe bias in the Alps region. Gradual improvements are seen with simple fusion, location embeddings, and loss regularization, with the greatest improvement observed through embeddings. Furthermore, we present a regularization framework that demonstrates further improvements over the embedding-based method and potential for further development [18, 26].

2 Related work

2.1 Weather forecasting

The shift from GCM- and NWP-based weather modeling towards deep learning-based forecasting [4] has motivated the development of numerous AI-based strategies and frameworks. Several studies have compared proposed pipelines across different forecasting tasks — covering temporal, spatial, extremity, and parameter domains—have been conducted [27]. A variety of architectures have been proposed, which can be broadly classified into general-purpose vision deep learning models, including Recurrent Neural Networks, Temporal Convolutional Networks, and Vision Transformers [4, 28], and specialized architectures designed specifically for weather forecasting, such as FourCastNet [13, 29] and NeuralGCM [14]. Both the task formulation and architectural choice vary significantly depending on the specific forecasting objective. Generally, these pipelines are optimized using a data-driven objective, such as mean squared error (MSE) [27].

2.2 Geography-informed learning

Various strategies for geography-informed learning have been proposed. The strategies can be broadly grouped into three main directions:

- Simple fusion
- Geographical positioning encoding
- Informed optimization objective

Simple fusion. By simple fusion, I summarize the methods that append a raw prior to the input, which can be seen as the concatenation of the input tensor with the prior tensor containing geographical information. In contrast to more complex methods of utilizing the prior information, this approach is easy to implement and often serves as a benchmark for more refined methods. Simple fusion was, for example, used by Vandal et al. [30], who added topography as an additional channel to the task’s input.

Geographical positioning encoding Geographical positional encoding has been studied as a standalone approach [31, 32], or as an intermediate step for other downstream tasks such as downscaling¹ [20, 33]. Geographical positional encoding therefore uses an encoder framework that is either trained in an encoder–decoder style or jointly trained with the task’s objective to yield encodings of the geolocation, optimized for the respective task. Russwurm et al. [31] proposed an encoding strategy based on spherical harmonics and sinusoidal representation networks, which

¹Obtaining a higher resolution (superresolution) of the map of a parameter such as wind speed or temperature

have been shown to be effective in downscaling tasks when combined with frequency-aware input transformations [33].

Informed optimisation objective Another method to capitalize on a physics prior is to use a physics-constrained objective, i.e., to optimize with respect to a custom physics-derived loss function. [34] proposes different frameworks, such as loss-constrained and architecture-constrained settings. To give an example of such a setting, [35] jointly learns PDE parameters with an additional PDE-based loss term.

2.2.1) Research Gap

Although several advanced methods exist in the field of geography-informed learning (e.g., geographical location encoding [31, 33] or informed optimization objectives [34, 35]) for certain downscaling [33] and forecasting tasks [20, 27, 28], the impact of the topography prior on temperature forecasting has, to our knowledge, not yet been systematically studied. While many techniques provide useful building blocks for incorporating spatial or physical information into data-driven models, the explicit relationship between topography and temperature remains largely uninvestigated. In particular, there are no established methods that directly exploit the influence of terrain features (e.g., elevation, slope, coordinates) to improve temperature forecasting accuracy in complex terrain regions. This represents a significant gap, as topography strongly modulates surface temperature through altitude-dependent climate effects and local orographic influences, especially in mountainous areas. Hence, incorporating topographic priors could improve both the predictive accuracy and the physical consistency of deep learning-based temperature forecasting.

This project addresses this gap by exploring various approaches to integrate topographic and location² information into deep learning-based 2m temperature forecasting. In particular, we explore different ways in which topography can be leveraged for geography-informed learning by adapting forecasting neural network architectures for simple fusion and geographical positional encoding. In addition, we propose a novel empirical loss regularization framework that incorporates relationships between topographic priors and training data.

3 Methods

3.1 Definitions

3.1.1) Grid Variables

In order to set the task, we first define a grid variable.

DEFINITION 1 (GRID VARIABLE (GV)): Let $I \in \mathbb{R}^{P_x \times P_y}$ denote a *Grid variable*, representing the assignment of a value to each pixel (i, j) on a 2D grid of size $P_x \times P_y$. The value at each pixel is defined as the average of a continuous function $H(x, y)$ over the area of the pixel, i.e.,

$$I(i, j) = \frac{1}{A_p} \int_{A_p} H(x, y) dx dy, \quad (1)$$

where A_p is the area of a single pixel p .

The available training data (details in subsection 4.1.1) provide measurements of near-surface temperature at an altitude of 2m above ground, aggregated over 11 km² squares. Based on this, we can define the task-relevant grid variables.

²As topography is $h(x, y)$, adding topography implicitly or explicitly provides location information

DEFINITION 2 (2M-TEMPERATURE GV): Let *2m-temperature* $T_{2m}(i, j)$ be the *Grid variable* representing temperature at 2m above ground.

Further, we define orography on the same grid as the *2m-temperature* data:

DEFINITION 3 (OROGRAPHY GV): Let *Orography* $H(i, j)$ be the *Grid variable* representing altitude above sea level.

3.1.2) Grid Variable Kernels

Based on a given *Grid variable*, different local metrics can be computed by applying kernel filters [36] to yield new *Grid variables*. To obtain a suitable representation of the topography, simple and useful kernel filters are gradient approximations, such as those derived from first-order finite differences:

$$\nabla_{\mathbb{R}^2} H(x, y) = (\partial_x H, \partial_y H) \quad (2)$$

Equivalently, we can define this on the sphere using $\nabla_{\mathbb{S}^2} H(\lambda, \phi)$; however, for conceptual simplicity, the planar version is explained.

DEFINITION 4 (GRADIENT GRID VARIABLES): We define the grid variables *horizontal gradient* $\partial_x(i, j)$ and *vertical gradient* $\partial_y(i, j)$ as first-order forward-difference gradient approximations on a regular grid with uniform spacing:

$$\partial_x H(i, j) = \frac{H(i+1, j) - H(i, j)}{\Delta x}, \quad (3)$$

and $\partial_y H(i, j)$ analogously.

For computational efficiency and to avoid unnecessary padding, a forward difference can be used rather than, e.g., a central difference, and the grid scaling factor Δx can be omitted, yielding the approximation $\partial_x H(i, j) = H(i+1, j) - H(i, j)$. Again, $\partial_y H(i, j)$ is computed analogously. As a result, the computed gradients represent normalized local slope information rather than physically scaled derivatives. In the same spirit, one-sided padding of width *one* is applied at the domain boundaries to preserve the original grid dimensions.

3.1.3) Task description

We first provide a definition of the main task to be solved.

DEFINITION 5 (DIRECT FORECASTING): *Direct forecasting* is defined as the task of predicting a time step of a time series that lies Δt time steps in the future from h time steps sampled at an interval w backward from t_0 (i.e., $t_0, t_{-w}, \dots, t_{-hw}$), using a model M such that $\mathcal{T}_{pred} = M(\mathcal{T}_{in}, \theta)$, where

- *Input* $\mathcal{T}_{in}$: the input tensor to the model, of dimension $h \cdot C_{var}^{in} \times P_x \times P_y$,
- *Output* $\mathcal{T}_{out}$: the output tensor of the model, of dimension $C_{var}^{out} \times P_x \times P_y$,
- *History* h : the number of historical frames taken into consideration,
- *Window* w : the step size between considered historical frames,
- *Lead time* Δt : the interval indicating how far into the future the prediction is made.

Where P_x, P_y denote the height and width, and C_{var}^* the number of channels. Figure 2 visualizes the parameters *History*, *Window* and *Lead time*. The model parameters θ can then be optimized using an optimization algorithm in a supervised learning framework.

For this project, the task simplifies, as we have only one input and one output channel: *2mTemperature*, representing the temperature 2 meters above ground. Consequently,

$$C_{var}^{in} = C_{var}^{out} = 1. \quad (4)$$

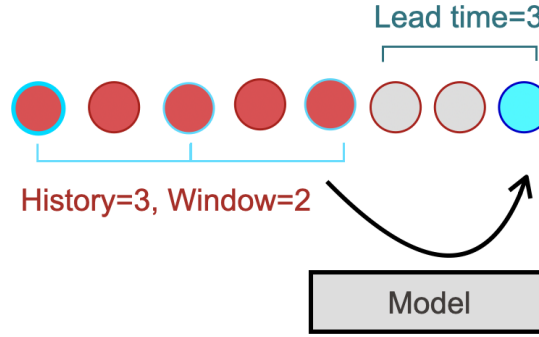


Figure 2: Visualization of the *History*, *Window*, and *Lead-time* parameters in the forecasting task

3.2 Baseline

As a baseline, the Vision Transformer (ViT) [37] implementation provided by Grover et al. via the `climate_learn` benchmark is used [4].

Essentially, the original ViT prediction head is modified to match the desired shape of $\mathcal{T}_{out}$. The *History*, *Lead time*, and *Window* parameters are taken to be identical to the default baselines provided by [4]. These parameters are summarized in Table 1, together with model-specific hyperparameters. Note that layer sizes and numbers are chosen to be as large as possible while still fitting on the available compute resources (10 GB GPU).

Forecasting Hyperparameters		Model Hyperparameters	
History h	3	Patch Size	6 ($66km \times 66km$)
Lead Time Δt	$2\mathcal{I}$ ($6h$)	Train Loss	MSE
Window w	$2\mathcal{I}$ ($6h$)	Eval Metric	RMSE

Table 1: Model configuration parameters. $\mathcal{I} = 3h$ denotes the interval between measurements in the time series.

3.3 Simple Fusion

One simple way of incorporating physical information into a model is to add it as an additional input channel. For forecasting methods, a common framework is to have an input domain with

spatial $S = (P_x, P_y)^3$, temporal T , and parameter (also referred to as channel) C_{var} dimensions. The overall data dimension is then $S \times T \times C_{var}$ ⁴.

Within this framework, one can treat an additional physical constraint—even if it is constant throughout the spatial or temporal dimensions (S, T)—as an additional parameter channel. This is logically equivalent to concatenating the existing parameter channels with the physical prior channels, resulting in a new data dimension $S \times T \times (C_{var} + 1)$ [30].

Fusion can be further enhanced by mixing information of the channels with the prior pixel-wise, which can define new composite channels such as:

$$C_i \cdot P + P, \quad \forall i \in [\dim(C_{var})] \quad (5)$$

where the operations are made pixelwise, C_i denotes the i^{th} channel and P denotes a grid variable representing a physical prior.

3.4 (geographical) Implicit Neural Representation - gINR

The general idea of gINR is to learn a positional encoding $\mathbf{PE}(x) : \mathbb{R}^{1 \times P_x \times P_y} \rightarrow \mathbb{R}^{N_{coeff} \times P_x \times P_y}$, based on *Orography*, which is a function of the location. Note that the spatial dimensions are either given implicitly by the grid or specified explicitly for each pixel on the grid. Then the channel dimension of the input to PE would be 3 rather than 1.

3.4.1) GeoFAR-style gINR

The authors of the GeoFAR paper [33] presented a Positional Encoding scheme that proved effective for downscaling tasks. Extracting the relevant blocks from their pipeline yields a candidate Positional Encoding for forecasting.

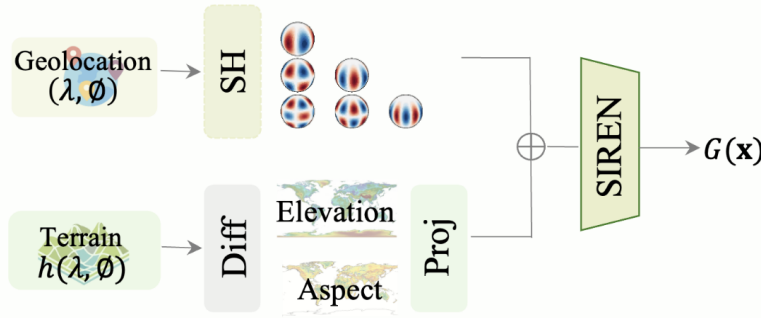


Figure 3: Pipeline of the gINR framework for geographical positional encoding with sinusoidal-harmonics terrain encoding and SIREN encoder.

The gINR consists of a geolocation encoding through Spherical Harmonics as well as an encoding of the terrain vector consisting of altitudes and gradients. Both encodings are fused using a Sinusoidal Representation Network (SIREN) [38] to yield the final positional encoding (see Figure 3 [33]).

We formally define these stepwise mappings:

³for $\dim(S) = 2$; if $\dim(S) = 3$, one has $S = (P_x, P_y, P_z)$

⁴Often, the temporal and parameter dimensions are stacked, and this new dimension of $T \times C_{var}$ is then referred to as input channels C_{in} (especially when $T = 1$ or $C_{var} = 1$). Common notations for the total data dimension (with $\dim(S) = 2$) are $C_{in} \times P_x \times P_y$ or $\dim(T) \cdot C_{var}^{in} \times P_x \times P_y$.

DEFINITION 6 (SPHERICAL LOCATION ENCODING): We define the *Spherical location encoding* as the encoding of longitude and latitude (λ, ϕ) using a band-limited spectral harmonic expansion. Formally, let $\{Y_l^m\}_{l \geq 0, |m| \leq l}$ denote the real spectral harmonic basis (with l : degree, m : order), orthonormal on $\mathbb{S}^2$: $\int_{\mathbb{S}^2} Y_l^m(\omega) Y_{l'}^{m'}(\omega) d\omega = \delta_{ll'} \delta_{mm'}$. Truncating at $l = L$ yields a multi-scale and rotation-aware encoding of dimension $C_{sh} = (L+1)^2$, resulting in an intermediate positional encoding of dimension $\mathbb{R}^{C_{sh} \times P_x \times P_y}$, with parameter $\sqrt{C_{sh}} = L+1$ defining its size [33]:

$$\mathbf{Y}_{\mathbf{C}_{sh}}(x) \in \mathbb{R}^{C_{sh} \times P_x \times P_y} = \mathbf{Y}_{\sqrt{C_{sh}}-1}^{\mathbf{P}}, \quad \forall p \in x, \quad (6)$$

where $\mathbf{Y}_{\mathbf{L}}^{\mathbf{P}}(\lambda, \phi) = [Y_l^m(\lambda, \phi)]_{l=0 \dots L, m=-l \dots l} \in \mathbb{R}^{C_{sh}}$.

DEFINITION 7 (TERRAIN VECTOR): Define the *Terrain vector* as $\mathbf{T} = [H, \partial_x H, \partial_y H]$. It's encoding, $\Psi(\mathbf{T})$, maps into $\mathbb{R}^{C_{terr}}$, resulting in a grid variable of dimension $\mathbb{R}^{C_{terr} \times P_x \times P_y}$ as follows [33]:

$$\Psi_{C_{terr}}(\mathbf{T}, \theta) : \mathbb{R}^{3 \times P_x \times P_y} \rightarrow \mathbb{R}^{C_{terr} \times P_x \times P_y} = \text{Conv2D}_{k=3 \times 3}(\mathbf{T}), \quad (7)$$

where $\text{Conv2D}_{3 \times 3}(\cdot)$ denotes a 2D convolutional neural network with kernel size 3×3 . The final positional encoding for the implicit neural representation is obtained by choosing C_{sh} as the square of an integer and setting $C_{terr} = C_{sh}$. The encoding then consists of passing the sum of the terrain and location encodings to a sinusoidal representation network (SIREN) [38]. The authors of the GeoFAR paper [33] justifies this choice based on its capability to model fine details, resulting in [33]:

$$\mathbf{PE}(x) = \text{NN}_{\text{SIREN}}(\mathbf{Y}_{\mathbf{C}_{sh}} + \Psi_{C_{sh}}(\mathbf{T})) \quad (8)$$

3.5 Explicit Empirical Consistency Regularization (EECR)

Many physics-informed learning approaches enforce known deterministic laws by constraining model predictions to satisfy governing equations, often expressed as partial differential equations (PDEs, e.g., Navier-Stokes) which govern atmospheric flows. In the context of near-surface temperature forecasting, however, no explicit governing equation directly relates temperature fields to complex orography at the spatial and temporal scales considered. While approximate physical relationships exist (e.g., adiabatic lapse rates), these are insufficient to define a closed-form constraint due to the problem's complexity.

We therefore introduce *Explicit Empirical Consistency Regularization (EECR)*, a general framework that incorporates physical intuition through structural consistency constraints rather than explicit equation enforcement. The main idea is to regularize model predictions so that the spatial structures induced by terrain in the predicted *2m-temperature* field are statistically consistent with empirically observed terrain-temperature relationships.

EECR framework We first define the general framework such that the EECR term can be chosen as a specific instance of it. We start by introducing definitions and notation.

DEFINITION 8 (TENSOR SPACE NOTATION): Let D denote a physical domain (e.g., orography H or temperature T); then the notation

$$\mathcal{T}^\alpha(\mathbb{R}_D)$$

defines the space of real-valued tensors of shape α , whose entries carry the physical units associated with the domain D . *Notation detail:* We reserve subscripts for the dimensions of physical domains and superscripts for the spatial dimensions of tensors.

DEFINITION 9 (SPATIAL OPERATOR): A mapping

$$\phi_D : \mathcal{T}^{\text{in}}(\mathbb{R}_D) \rightarrow \mathcal{T}^{\text{target}}(\mathbb{R}_D)$$

is called a *spatial operator* if it aggregates (or extrapolates) spatial information from $\alpha = \text{in}$ to $\alpha = \text{target}$ while preserving the physical units of domain D in each output entry.

Typical examples include convolutions (e.g., finite-difference kernels), grid-partition-wise statistics (e.g., patch summaries), or global summaries. The aggregation (or extrapolation) shape *target* may differ from the original spatial resolution (usually $\text{in} = P_x \times P_y$).

DEFINITION 10 (DOMAIN MAPPING): Given physical domains D and G , a mapping

$$\psi_{D \rightarrow G} : \mathcal{T}^\alpha(\mathbb{R}_D) \rightarrow \mathcal{T}^\alpha(\mathbb{R}_G)$$

is called a *domain mapping* if it acts identically at every spatial location and maps quantities from domain D to domain G .

DEFINITION 11 (AGGREGATE CONSISTENCY LOSS): Let ϕ_H and ϕ_T be *spatial operators* applied to the *orography* $H(i, j)$ and the predicted temperature field $\hat{T}$, respectively, and let $\psi_{H \rightarrow T}$ be a *domain mapping*. Let α denote the shape of the output tensors of $\phi_T(\hat{T})$ and $\psi_{H \rightarrow T}(\phi_H(H))$, and let $p \in \alpha$ index individual entries. Then, the aggregate consistency loss at entry p is defined as

$$\mathcal{L}_{\text{ACL}}(p) = \mathcal{L}_{\text{metric}}\left(\psi_{H \rightarrow T}(\phi_H(H))_p, \phi_T(\hat{T})_p\right), \quad (9)$$

where $\mathcal{L}_{\text{metric}}$ is a suitable pointwise metric (e.g., MSE or MAE).

DEFINITION 12 (EECR LOSS CONTRIBUTION): The *EECR Loss contribution* is then obtained by evaluating $\mathcal{L}_{\text{ACL}}(p)$ ⁵, at each entry $p \in \alpha$ and summing over all entries:

$$\mathcal{L}_{\text{EECR}} = \sum_{p \in \alpha} \mathcal{L}_{\text{ACL}}(p). \quad (10)$$

Provided that ϕ , ψ , and $\mathcal{L}_{\text{metric}}$ are differentiable, this regularization is compatible with gradient-based optimization.

3.5.1) Histogram-based EECR

We now instantiate the general framework with specific operator choices. The histogram-based EECR aims to empirically capture the (dataset-specific) relationship between terrain gradients and the resulting temperature differences. To do this, we first compute local orographic differences ($|\Delta H|$) using a discrete gradient proxy. We then partition the range of $|\Delta H|$ into discrete bins $\{\mathcal{B}_k\}_{k=1}^K$. For each bin, we compute the average absolute temperature difference ($|\Delta T|$) observed in the training data, giving a conditional expectation:

$$\hat{g}(b_k) = \mathbb{E} \left[|\Delta T| \mid |\Delta H| \in \mathcal{B}_k \right] \quad (11)$$

This provides a stepwise, empirical mapping from terrain gradient magnitudes to expected temperature differences. To make this mapping differentiable for use in training, it is smoothed using curve fitting, resulting in a continuous domain mapping $\psi_{H \rightarrow T}$.

⁵Typically with the mean squared error (MSE) as pointwise metric.

Intuitively, this mapping defines the expected temperature differences for all terrain gradients. The EECR loss penalizes deviations of the predicted temperature gradients from this empirical expectation. In practice, small terrain gradients are effectively zeroed out using a rescaled sigmoid, reflecting the assumption that these gentle slopes are already well captured by the model. Consequently, the regularization primarily targets larger gradients, where terrain-induced temperature variations are more difficult to learn. We formally define the *spatial operator* and the *domain mapping* as follows:

Spatial operator We first define a suitable spatial operator. A simple approach is to compute vertical and horizontal neighbor differences and concatenate them. The underlying intuition is that topography differences (i.e., gradients) naturally induce temperature variations.

DEFINITION 13 (DISCRETE ABSOLUTE GRADIENT PROXY): For a grid variable $X \in \mathbb{R}^{P_x \times P_y}$, the *discrete absolute gradient proxy* is defined according to Def. 4 as

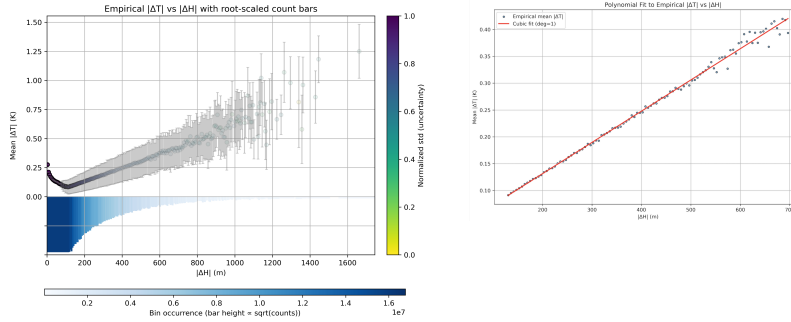
$$|\widehat{\nabla}|(X) = \left(|X_{i+1,j} - X_{i,j}|, |X_{i,j+1} - X_{i,j}| \right) \in \mathbb{R}^{2 \times (P_x - 1) \times (P_y - 1)}. \quad (12)$$

The discrete gradient proxy is applied identically to orography and temperature fields:

$$\phi_H = |\widehat{\nabla}|(H), \quad \phi_T = |\widehat{\nabla}|(T). \quad (13)$$

Domain mapping Let $\{\mathcal{B}_k\}_{k=1}^K$ be a partition of the range of $|\Delta H|$ into discrete bins. The histogram-based domain mapping $\psi_{H \rightarrow T}$ is first defined via the conditional expectation (Equation 11) and then extended to a continuous function using curve fitting:

$$\psi_{H \rightarrow T}(|\widehat{\nabla}|(H)_p) \approx \hat{g}(|\Delta H|). \quad (14)$$



(a) Histogram plot of binned $|\Delta T|$ data (b) Linear fit for $|\Delta H| \geq 130m$

Figure 4: Histogram plots of $|\Delta T|$ in $|\Delta H|$ -bins and curve fitting.

After inspection of the histogram plot (see Figure 4a), the following curve-fitting strategy is applied: in the high-gradient regime, a linear fit is used to obtain the parameters (see Figure 4b); a rescaled sigmoid function $\hat{\sigma}$ is used to zero out the loss in lower-gradient regimes where simple linearity does not apply:

$$\psi_{H \rightarrow T}(|\widehat{\nabla}|(H)_p) = \hat{\sigma} \left(0.0005816 |\widehat{\nabla}|(H)_p + 0.01518 \right), \quad \hat{\sigma}(x) = \sigma \left(\frac{x - 130}{20} \right), \quad (15)$$

where σ is the standard sigmoid function. This zeroing-out is motivated by the fact that in low- H regimes, the variance of temperature differences is small, meaning the forecasting model can learn these patterns easily; therefore, regularization in this regime is less critical.

Using mean squared error as the aggregate consistency loss, the resulting EECR term is

$$\mathcal{L}_{\text{HIST}} = \frac{1}{2} \left(\hat{\sigma} (0.0005816 |\widehat{\nabla}|(H) + 0.01518) - |\widehat{\nabla}|(\hat{T}) \right)^2. \quad (16)$$

3.5.2) Space-Dependent EECR

Using the same *spatial operator* as previously defined (see Definition 13), we refine the domain mapping. The intuition is that the orography–temperature relationship may vary across locations, so a parametric, space-agnostic mapping can learn this relationship while remaining translation-invariant.

DEFINITION 14 (LEARNED DOMAIN MAPPING): Denote a *learned domain mapping* with learnable parameters θ as $\psi_{H \rightarrow T}^\theta$.

For space-agnostic EECR, we specify the learned model as a shallow 2D convolution applied identically across spatial locations:

$$\psi_{H \rightarrow T}^\theta = \text{Conv2D}_{3 \times 3}(\cdot), \quad (17)$$

where as in Equation 7, $\text{Conv2D}_{3 \times 3}(\cdot)$ denotes a 2D convolutional neural network with kernel size 3×3 . The corresponding aggregate consistency loss $\mathcal{L}_{\text{ACL}}$ (see Equation 9) is:

$$\mathcal{L}_{\text{SD}} = \frac{1}{2} \left(\text{Conv2D}_{3 \times 3}(|\widehat{\nabla}|(H)) - |\widehat{\nabla}|(\hat{T}) \right)^2. \quad (18)$$

3.5.3) Geography-Aware Space-Dependent EECR

Going one step further, one can introduce a gINR-style geography encoding layer before the convolution, yielding:

$$\psi_{H \rightarrow T}^\theta = \text{Conv2D}_{3 \times 3}(\mathbf{PE}(\cdot)),$$

where $\text{Conv2D}_{3 \times 3}(\cdot)$ is defined as above, and $\mathbf{PE}(x)$ is the geography encoding described in 3.4.1). The ACL $\mathcal{L}_{\text{ACL}}$ is (analogously to 3.5.2):

$$\mathcal{L}_{\text{GSD}} = \frac{1}{2} \left(\text{Conv2D}_{3 \times 3}(\mathbf{PE}(|\widehat{\nabla}|(H))) - |\widehat{\nabla}|(\hat{T}) \right)^2. \quad (19)$$

Adding positional encoding allows the mapping to vary smoothly with geography while remaining structurally constrained by local terrain gradients.

4 Experiments

4.1 Experimental Setup

4.1.1) Data

The experiments are conducted on the Copernicus European Regional ReAnalysis (CERRA) dataset produced by C3S/ECMWF. The field of 2-m air temperature (t2m) was used as the single variable. The spatial resolution is 11 km, $(P_x, P_y) = (534, 534)^6$. The data consists of 3-hourly measurements over the time span from 2018 to 2021, yielding 2920 time frames per year [10, 25]. The Train/Test split was done as follows: Train: 2018–2020 (8760 frames) and Test: 2021 (2920 frames).

⁶The native data had a 5.5 km resolution (1069×1069), and the resolution used here was obtained through bilinear interpolation by [33].

4.1.2) Implementation Details

Training and Evaluation Training was performed on an NVIDIA GeForce GTX TITAN X GPU with 12 GB of VRAM. Experiments were conducted with a batch size of 16, for 20 epochs, and with early stopping⁷. The model and task-relevant hyperparameters were chosen as described in Section 3.2.

DEFINITION 15 (RMSE): The Root Mean Squared Error (RMSE) is defined as:

$$RMSE = \sqrt{\frac{1}{P_x P_y N} \sum_{n \in \mathcal{D}} \sum_{i=1}^{P_x} \sum_{j=1}^{P_y} \left(\hat{T}_{2m_n}(i, j) - T_{2m_n}(i, j) \right)^2} \quad (20)$$

and was used as the validation metric, where n indexes a sample in the test or validation set $\mathcal{D}$. Additionally, we define the pixel-wise Mean Absolute Bias (MAB) as a metric to inspect regional biases:

DEFINITION 16 (PIXEL-WISE MEAN ABSOLUTE BIAS):

$$MAB(i, j) = \frac{1}{N} \sum_{n \in \mathcal{D}} \left| \hat{T}_{2m_n}(i, j) - T_{2m_n}(i, j) \right| \quad (21)$$

The global Mean Absolute Bias can then be computed as $MAB = \frac{1}{P_x P_y} \sum_{i,j} MAB(i, j)$, and was reported as an additional metric. Mean Squared Error (MSE) training loss was used throughout all experiments. The EECR regularizations (see subsection 3.5), given in Equations (16), (18), and (19), were tested with $\lambda = 0.1$ as additional constraints, according to the objective:

$$\mathcal{L} = \mathcal{L}_{MSE} + \lambda \mathcal{L}_{ACL} \quad (22)$$

Optimization To optimize the model with respect to the respective objective, we used the Adam optimizer [39] with a learning rate of 5×10^{-4} , weight decay 1×10^{-5} , and exponential decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.99$ for the first and second moments, as used by [4] for their baselines.

Codebase Grover et al. [27] provide a PyTorch-based library, **ClimateLearn**, which implements a range of functionalities for benchmarking ML models for weather and climate, providing baselines across different forecasting models. We extended the **ClimateLearn** library [4] using PyTorch [40] to support all methods described in Section 3, including the necessary extensions for model-specific features (e.g., custom arguments, metrics, visualizations). This allows **ClimateLearn**’s native setup to be used for model training. The preprocessing, experiments, and visualization scripts were then built using the extended library `climate_learn_oro`.

4.2 Results

4.2.1) Main Comparison

Table 2 reports performance on RMSE and MAB (defined in 4.1.2) for the evaluated models and configurations. We note substantial improvements of simple fusion over no prior, as well as the benefits of geography encoding (via encoder or EECR) over simple fusion. The best result was obtained using EECR via the histogram approach.

⁷If there is no decrease in training loss for 5 epochs.

Table 2: Test and validation Root Mean Square Error (RMSE) and Mean Absolute Bias (MAB) for the evaluated models. Lower values indicate better performance. The table is ordered by decreasing RMSE (indicated by the symbol $\downarrow$). Model and EECR abbreviations follow the naming conventions defined in Table 3 and Table 4 respectively, found in Appendix A.1 and A.2.

Model	Train Loss	EECR($\lambda = 0.1$)	RMSE $\downarrow$	MAB
ViTgINR	MSE	HIST	2.4662	1.749
ViTgINR	MSE	Conv/PE	2.4718	1.756
ViTgINR	MSE	Conv	2.4795	1.761
ViTgINR	MSE	-	2.5028	1.776
ViT (Baseline)	MSE	HIST	2.7415	1.980
ViTCC	MSE	-	2.7507	1.980
ViT (Baseline)	MSE	-	2.7910	1.982

4.2.2) Bias Analysis

In addition to simple metric reporting, it is important to inspect local biases to assess whether the topography prior indeed improves predictions in areas where topography changes rapidly, and predictions are therefore expected to be more difficult. In Figure 5, we report the pixel-wise Mean Absolute Bias (MAB) over the predictions.

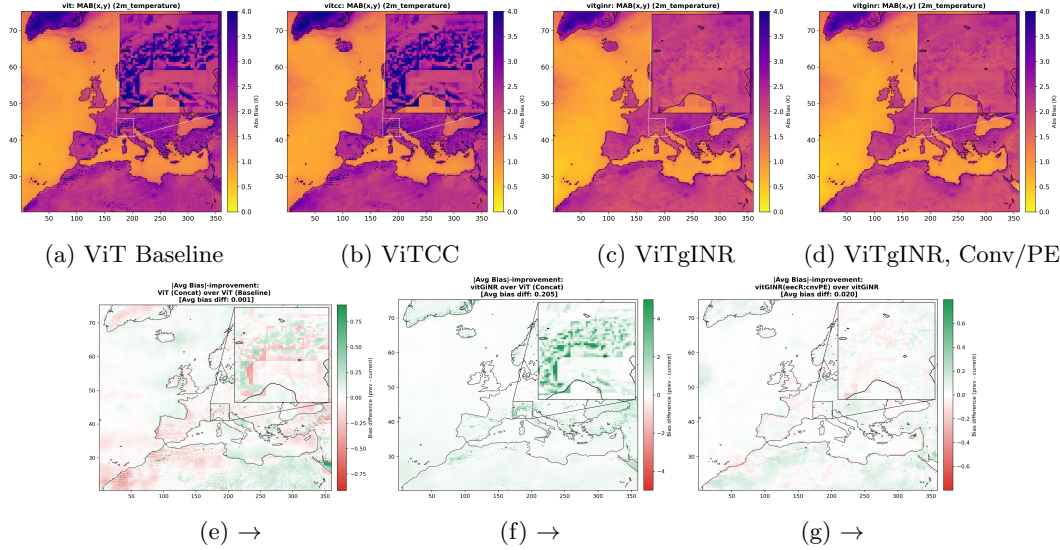


Figure 5: $MAB(i, j)$ for selected models and their comparisons. Subfigures (e)–(g) show the relative improvement of models (a)–(d) in sequential comparisons.

The top row of Figure 5 presents the absolute biases according to the $MAB(x, y)$ metric (Equation 21). The bottom row compares the relative differences in bias between the adjacently displayed models. Inspection of these maps confirms the initial observation: there is a large gap from baseline and concat to the encoding model, while relatively small deviations exist between the different versions of the respective model and loss modifications. Furthermore, substantial improvements—especially in the region of the Alps—are observed in all proposed models, with significant reductions in bias magnitude.

4.3 Discussion

4.3.1) General Observations

Land–Sea Gap We observe a clear prediction difference in the bias between land and sea pixels (Figure 5). A possible explanation is the comparatively low variance of sea surface temperature, as water bodies may dampen atmospheric effects due to their heat capacity. This can also be related to topography: there are no local topographic differences over the sea, and therefore we do not expect sharp temperature changes, which mainly stem from atmospheric effects. Notably, this highlights the importance of terrain-informed encoding in improving predictions over land regions where topography varies rapidly.

Greenland Anomaly Greenland exhibits a high bias overall across all models (see Figure 5). This may be due to the fact that temperatures in Greenland are relatively low compared to the rest of the CERRA region, causing the models to be biased toward predicting higher temperatures in general. Slight increases over Siberia further support this hypothesis. This observation suggests that models may benefit from additional normalization or region-specific calibration, particularly in extreme climate regions. Another hypothesis is that the anomaly may result from the distortion of the sphere shape. As deep learning methods mostly use rectangular grids, high-latitude regions often exhibit some degree of distortion [41] which could be amplified due to more complex terrain in Greenland.

PE Improvement Significantly larger and less consistent biases are observed for models without positional encodings (Figure 5a and 5b). Using the ViT backbone on positional encodings rather than raw inputs improves prediction consistency. This confirms that the gINR-style positional encodings and EECR regularization not only improve accuracy but also enhance the structural consistency of predictions across spatial locations. Furthermore, discontinuous predictions can appear over patch borders of the ViT. This can be explained by a well-known characteristic of transformers: they require a large amount of data to work effectively.

Divergence of MAB and RMSE For some model comparisons (e.g., ViT (Figure 5a) vs. ViT Concat (Figure 5b), or different EECR terms (Figure 5c and 5d), we note that while RMSE may indicate considerably better performance for one model, the pixel-wise MAB (x, y) difference plots show minimal changes. This is also reflected in the total MAB. Such divergence can result from outlier sensitivity in RMSE, which may overemphasize a few extreme errors while leaving regional biases largely unchanged. Hence, combining RMSE with MAB provides a more comprehensive assessment of both global accuracy and local bias mitigation, demonstrating the effectiveness of EECR in reducing structured errors.

4.3.2) Model Comparisons

Simple Fusion We first note that while simple concatenation slightly improves the RMSE (Table 2), it does not clearly reduce the bias in the Alpine region (Figure 5b), which is a particularly challenging area to predict. Several factors could explain this phenomenon. When simply concatenating topography, the model must first learn the complex relations between modalities, and it has no explicit indication of the different data types. Therefore, longer training may be necessary, or the model might not be powerful enough to capture the required connections. The modest improvement in RMSE may reflect that the prior provides some guidance, or that slight changes in the architecture—even with a similar number of parameters—can produce better predictions.

(geographical) Implicit Neural Representation The largest reduction in prediction bias is achieved by applying positional encoding via the gINR framework (see Figure 5f, and Figure 7). Both the encoding itself and the geography-awareness it introduces can explain this improvement, as the model is now able to leverage spatially structured information effectively.

EECR We first observe that adding EECR slightly improves RMSE over the standard ViT-gINR (see Table 2), indicating that the chosen spatial operator, enforcing $|\Delta H| - |\Delta T|$ consistency, is indeed helpful when applied as a regularization term. However, inspection of the bias plots (Figure 5g respectively Figure 5c vs. 5d) does not reveal clear improvements in mountainous regions, making it difficult to draw definitive conclusions. We also note that the heuristic histogram approach yields slightly better results, which may be explained by its more general regulation, targeting primarily areas with large differences due to weighting.

While the experiments progressively increase architectural and constraint complexity, it remains important to verify whether EECR itself is effective by applying the regularization term to the baseline model.

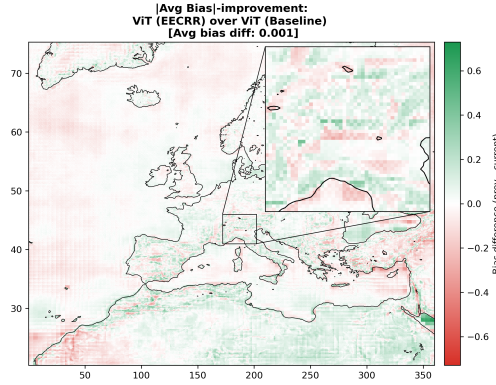


Figure 6: Comparison of MSE and MSE + EECR ($\lambda = 0.1$) objectives.

From Figure 6, we observe that solely injecting positional encoding via EECR performs better than the baseline. However, comparing Figure 5 with Figure 6 shows worse performance of ViT with EECR than ViTgINR alone (bias decrease on the order of 0.5 compared to 2–3 for ViTgINR). Nevertheless, the method as a whole still provides measurable improvements over the baseline, suggesting that further exploration of EECR—through fine-tuning (note that $\lambda = 0.1$ may be too low) or alternative choices of spatial operator and domain mapping—can be beneficial.

5 Conclusion

5.1 Summary

The achievements of this project can be summarized as follows:

- We implemented a range of methods to integrate topography priors into neural network frameworks, exploring different ways to encode and fuse geographic information with the forecasting models.
- The forecasting performance of these methods was systematically evaluated both quantitatively and qualitatively, allowing us to identify the strengths and weaknesses of each approach and assess the impact of topography-informed inputs.

- Building on these insights, we introduced an extendable framework for Explicit Empirical Consistency Regularization (EECR), which leverages spatial operators and domain mappings to enforce terrain–temperature consistency. We demonstrated that applying EECR can lead to modest improvements, highlighting its potential as a promising regularization framework for incorporating topography priors in forecasting tasks.

5.2 Outlook

5.2.1) Geography-informed Learning

For all categories of geography-informed learning, further approaches beyond those explored in this work could be investigated. Alternative simple fusion strategies, different encoding frameworks, and various objective constraints may provide additional performance gains.

More advanced encoding and modeling frameworks, such as *learnable Fourier features and Fourier Neural Operators (FNOs)* and [42, 43], *graph-based terrain embeddings* [44, 45] have the potential to better capture complex spatial and geographic relationships. These approaches can model both local and global structure, incorporate higher-order interactions, and generalize across irregular grids or resolutions.

Incorporating additional input modalities, such as land-cover type, vegetation indices, or satellite-derived albedo, could further enhance spatially-aware predictions and improve model generalization across diverse terrains. Multi-scale aggregation and attention mechanisms, commonly used in remote sensing applications [46, 47], provide flexible ways to combine information at different spatial scales, enabling models to capture fine-grained terrain variations while retaining broader contextual understanding.

Finally, replacing or augmenting static convolutional operators in encoding pipelines via attention-based or transformer modules may allow the model to flexibly encode complex terrain–temperature relationships, providing additional expressive power while maintaining structural constraints.

5.2.2) Extension of the EECR Framework

The Explicit Empirical Consistency Regularization (EECR) framework admits a wide range of natural extensions beyond the specific instantiations explored here. At the level of *spatial operators*, higher-order finite-difference operators, laplacians, curvature-based metrics, or multi-scale aggregations could be employed to capture local, regional, and scale-dependent terrain effects [43, 44, 46]. Such operators are well established in graph-based learning, neural operator theory, and remote sensing, where they enable physically meaningful inductive biases across irregular or multi-resolution spatial domains [43, 48].

Similarly, the *domain mapping* may be generalized from linear or convolutional forms to nonlinear parametric mappings, lightweight multilayer perceptrons, attention-based modules, or hybrid combinations thereof, allowing the model to adaptively weight spatial interactions and contextual dependencies beyond fixed local receptive fields [42, 47].

Furthermore, the regularization weight λ could be adapted spatially or temporally, either through hard coded, heuristics or via learnable mechanisms, prioritizing high-error regions or periods where topography has a stronger influence, which could improve both stability and predictive accuracy [43, 46].

5.3 Limitations

Evaluation All methods could have been tested more exhaustively to obtain a better estimate of the variance in model outputs, which would improve the ability to distinguish significant effects from noise. In particular, performing multiple runs for variance estimation, or evaluating the models across different years or geographical regions, would help to assess generalization and robustness.

Fine-tuning Hyperparameter tuning for non-architecture-critical parameters was omitted due to compute restrictions. Further fine-tuning could have revealed the full potential of the tested methods. Optimizing parameters such as learning rate schedules, batch sizes, or regularization weights (λ for EECR), could further enhance predictive accuracy and stability.

Alternative Methods Additional settings could have been tested for the described methods, and other methods not explored in this study could also be evaluated. The experiments presented here represent a subset of possible configurations. A full survey of approaches for incorporating topography priors in temperature forecasting – including but not limited to the frameworks mentioned above (5.2.1) – remains an important direction for future work.

6 Code availability

The `climate_learn_oro` package extension is available on GitHub at: github.com/janzzraggen/climate_learn_oro. The code for the experiments is available via the repository `CerrOroCast` on GitHub at: github.com/janzzraggen/CerrOroCast

7 Acknowledgments

I would like to thank Prof. D. Tuia for his insightful feedback during our midterm discussion. I also acknowledge the contribution of C. Xu and thank him for his guidance on the project, directive inputs, and valuable insights from his work on the GeoFAR framework [33]. I acknowledge the use of large language models (ChatGPT) for debugging assistance during implementation and for typo detection and L^AT_EX formatting during the writing of this report.

References

- [1] M. Rodell and B. Li, “Changing intensity of hydroclimatic extreme events revealed by GRACE and GRACE-FO,” *Nature Water*, pp. 1–8, 2023.
- [2] E. Coughlan de Perez, H. Ganapathi, G. I. T. Masukwedza, T. Griffin, and T. Kelder, “Potential for surprising heat and drought events in wheat-producing regions of USA and china,” *npj Climate and Atmospheric Science*, vol. 6, no. 1, p. 56, 2023.
- [3] P. Stott, “How climate change affects extreme weather events,” *Science*, vol. 352, no. 6293, pp. 1517–1518, 2016.
- [4] T. Nguyen, J. Jewik, H. Bansal, P. Sharma, and A. Grover, “ClimateLearn: Benchmarking machine learning for weather and climate modeling,” in *Advances in Neural Information Processing Systems*, 2023, datasets and Benchmarks Track. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/ed73c36e771881b232ef35fa3a1dec14-Abstract-Datasets_and_Benchmarks.html
- [5] G. Brunet *et al.*, “Advancing weather and climate forecasting for our changing world,” *Bulletin of the American Meteorological Society*, vol. 104, pp. E909–E927, 2023.
- [6] Y. Wu and W. Xue, “Data-driven weather forecasting and climate modeling from the perspective of development,” *Atmosphere*, vol. 15, p. 689, 2024.
- [7] European Commission, “About copernicus,” 2025, accessed 2025-11-10. [Online]. Available: <https://www.copernicus.eu/en/about-copernicus>
- [8] NASA, “NASA Earthdata: Open access for open science,” 2025, accessed 2025-11-10. [Online]. Available: <https://www.earthdata.nasa.gov/>
- [9] V. Eyring, S. Bony, G. A. Meehl *et al.*, “Overview of the coupled model intercomparison project phase 6 (CMIP6) experimental design and organization,” *Geoscientific Model Development*, vol. 9, no. 5, pp. 1937–1958, 2016.
- [10] S. Schimanke, M. Ridal, P. Le Moigne *et al.*, “CERRA sub-daily regional reanalysis data for europe on single levels from 1984 to present,” Copernicus Climate Change Service Climate Data Store, 2021, accessed 2025-11-04.
- [11] H. Zhang, Y. Liu, C. Zhang, and N. Li, “Machine learning methods for weather forecasting: A survey,” *Atmosphere*, vol. 16, p. 82, 2025.
- [12] I. Price, A. Sanchez-Gonzalez, F. Alet *et al.*, “Probabilistic weather forecasting with machine learning,” *Nature*, vol. 637, pp. 84–90, 2025.
- [13] T. Kurth, S. Subramanian, P. Harrington *et al.*, “FourCastNet: Accelerating global high-resolution weather forecasting using adaptive fourier neural operators,” *arXiv preprint arXiv:2208.05419*, 2022.
- [14] D. Kochkov, J. Yuval, I. Langmore *et al.*, “Neural general circulation models for weather and climate,” *Nature*, vol. 632, pp. 1060–1066, 2024.
- [15] F. Sioni, A. Manzato, G. Fasano, C. Lussana, and A. Pucillo, “On atmospheric pressure and temperature correlation across various terrain types,” *Atmospheric Research*, vol. 311, p. 107689, 2024.

- [16] L. M. J. Mahdi, M. H. Abood, and Z. T. Mohammed, “Statistical analysis of the correlation between land surface temperature and terrain elements in diyala governorate using remote sensing techniques,” *IOP Conference Series: Earth and Environmental Science*, vol. 1545, p. 012127, 2025.
- [17] “Examining the relationship between land surface temperature and landscape features using spectral indices with Google Earth Engine,” *Heliyon*, vol. 8, no. 9, p. e10668, 2022.
- [18] T. Haiden, I. Sandu, G. Balsamo, G. Arduini, and A. Beljaars, “Addressing biases in near-surface forecasts,” *ECMWF Newsletter*, no. 157, 2018, autumn 2018.
- [19] H. Zhang, Z. Pu, and X. Zhang, “Examination of errors in near-surface temperature and wind from WRF numerical simulations in regions of complex terrain,” *Weather and Forecasting*, vol. 28, pp. 893–914, 2013.
- [20] J. Lian, S. Huang, J. Shao *et al.*, “TerraWind: A deep learning-based near-surface winds downscaling model for complex terrain regions,” *Geophysical Research Letters*, 2024.
- [21] E. de Bézenac, A. Pajot, and P. Gallinari, “Deep learning for physical processes: Incorporating prior scientific knowledge,” *Journal of Statistical Mechanics: Theory and Experiment*, p. 124009, 2019.
- [22] K. E. Trenberth and D. J. Shea, “Relationships between precipitation and surface temperature,” *Geophysical Research Letters*, vol. 32, no. 14, p. L14703, 2005.
- [23] P. A. Schultz and M. S. Halpert, “Global analysis of the relationships among a vegetation index, precipitation and land surface temperature,” *International Journal of Remote Sensing*, vol. 16, no. 15, pp. 2755–2777, 1995.
- [24] M. Bhaud, J. H. Cha, J. C. Duchêne, and C. Nozais, “Influence of temperature on the marine fauna: What can be expected from a climatic change,” *Journal of Thermal Biology*, vol. 20, no. 1, pp. 91–104, 1995.
- [25] M. Ridal, E. Bazile, P. Le Moigne *et al.*, “CERRA, the copernicus european regional reanalysis system,” *Quarterly Journal of the Royal Meteorological Society*, vol. 150, no. 763, pp. 3385–3411, 2024.
- [26] Environmental Computational Science and Earth Observation Laboratory (ECEO), D. Tuia, and C. Xu, “Topography-aware weather forecasting in europe: A focus on heatwaves,” 2025, semester Project Proposal, École Polytechnique Fédérale de Lausanne (EPFL), ENAC IIE ECEO Laboratory, Sion, Switzerland.
- [27] J. Shi, A. Shirali, B. Jin *et al.*, “Deep learning and foundation models for weather prediction: A survey,” *arXiv preprint arXiv:2501.06907*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.06907>
- [28] M. Aracne, T. Ruga, C. Lops, D. Federico, L. Caroprese, E. Zumpano, S. Montelpare, M. Pierantozzi, F. Dattola, P. Iaquinta, M. Iusi, R. Greco, M. Talerico, V. Coscarella, L. Legato, I. Pellegrino, S. Bergamaschi, M. Orsini, R. Martoglia, A. Livaldi, A. Jelali, and S. Sbreglia, “Comparing deep learning approaches for weather forecasting: Insights from the precede project,” in *Proceedings of the Workshops of the EDBT/ICDT 2025 Joint Conference*, Barcelona, Spain, 2025, pp. 1–10, © 2025 Copyright for this paper by its authors. [Online]. Available: <https://ceur-ws.org/Vol-3946/DARLI-AP-9.pdf>

- [29] B. Bonev, T. Kurth, A. Mahesh *et al.*, “FourCastNet 3: A geometric approach to probabilistic machine-learning weather forecasting at scale,” *arXiv preprint arXiv:2507.12144*, 2025.
- [30] T. Vandal, E. Kodra, S. Ganguly *et al.*, “DeepSD: Generating high resolution climate change projections through single image super-resolution,” in *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1663–1672.
- [31] M. Rußwurm, K. Klemmer, E. Rolf, R. Zbinden, and D. Tuia, “Geographic location encoding with spherical harmonics and sinusoidal representation networks,” in *International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=PudduuffLa>
- [32] G. Mai, K. Janowicz, B. Yan *et al.*, “Multi-scale representation learning for spatial feature distributions using grid cells,” *International Conference on Learning Representations*, 2020. [Online]. Available: <https://openreview.net/forum?id=rJljd4KDH>
- [33] Anonymous, “GeoFAR: Geography-informed frequency-aware super-resolution for climate data,” in *ICLR Conference Submission*, 2025, openReview submission #3415. [Online]. Available: <https://openreview.net/forum?id=0WHpOekph0>
- [34] F. Zanetta, D. Nerini, T. Beucler, and M. A. Liniger, “Physics-constrained deep learning postprocessing of temperature and humidity,” *Journal of Atmospheric and Climate Science*, 2023. [Online]. Available: <https://journals.sagepub.com/doi/10.1177/23028982221114189>
- [35] Y. Luo, S. Fang, B. Wu, Q. Wen, and L. Sun, “Physics-guided learning of meteorological dynamics for weather downscaling and forecasting,” *arXiv preprint arXiv:2505.14555*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.14555>
- [36] A. Torralba, P. Isola, and W. T. Freeman, *Foundations of computer vision*. MIT Press, 2024.
- [37] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations*, 2021. [Online]. Available: <https://openreview.net/forum?id=YicbFdNTTy>
- [38] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, and G. Wetzstein, “Implicit neural representations with periodic activation functions,” in *Advances in Neural Information Processing Systems*, 2020.
- [39] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.
- [40] A. Paszke, S. Gross, F. Massa *et al.*, “PyTorch: An imperative style, high-performance deep learning library,” *arXiv preprint arXiv:1912.01703*, 2019. [Online]. Available: <https://arxiv.org/abs/1912.01703>
- [41] H. Weller, J. Thuburn, and C. J. Cotter, “Computational modes and grid imprinting on five quasi-uniform spherical C grids,” *Monthly Weather Review*, vol. 140, no. 8, pp. 2734–2755, 2012.
- [42] Z. Li, N. Kovachki, K. Azizzadenesheli *et al.*, “Fourier neural operator for parametric partial differential equations,” *arXiv preprint arXiv:2010.08895*, 2020. [Online]. Available: <https://arxiv.org/abs/2010.08895>

- [43] N. Kovachki, Z. Li, B. Liu *et al.*, “Neural operator: Learning maps between function spaces with applications to PDEs,” *Journal of Machine Learning Research*, vol. 24, pp. 1–97, 2023.
- [44] Z. Li, N. Kovachki, K. Azizzadenesheli *et al.*, “Neural operator: Graph kernel network for partial differential equations,” *arXiv preprint arXiv:2003.03485*, 2020. [Online]. Available: <https://arxiv.org/abs/2003.03485>
- [45] S. Geisler, A. Kosmala, D. Herbst, and S. Günnemann, “Spatio-spectral graph neural networks,” *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: <https://arxiv.org/abs/2405.19121>
- [46] Z. Mao, X. Tong, Z. Luo, and H. Zhang, “MFATNet: Multi-scale feature aggregation via transformer for remote sensing image change detection,” *Remote Sensing*, vol. 14, no. 21, p. 5379, 2022.
- [47] M. Koh, B.-C. Park, and S.-W. Lee, “Integrating locality-aware attention with transformers for general geometry PDEs,” *arXiv preprint arXiv:2504.13480*, 2025. [Online]. Available: <https://arxiv.org/abs/2504.13480>
- [48] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” *IEEE Signal Processing Magazine*, vol. 38, no. 6, pp. 18–42, 2021.

A Appendix

A.1 Model naming conventions

Table 3: Overview of model naming conventions used throughout this work, with references to the corresponding method descriptions.

Abbreviation	Name	Category	Ref.
ViT	V ision T ransformer	Baseline	(3.2).
ViTCC	C oncatenation	Simple Fusion	(3.3).
ViTgINR	(g eographical) I mplicit N eural R epresentation	Positional Encoding	(3.4).

A.2 Regularization naming conventions

Table 4: Overview of EECR naming conventions used throughout this work, with references to the corresponding method descriptions.

Abbreviation	Name	Category	Ref.
HIST	H istogram	Heuristic	(3.5.1).
Conv	C onvolution	Space-Aware	(3.5.2).
Conv/PE	C onvolution w/ P ositional E ncoding	Geo & Space-Aware	(3.5.3).

A.3 Validation metric plot for reported experiments

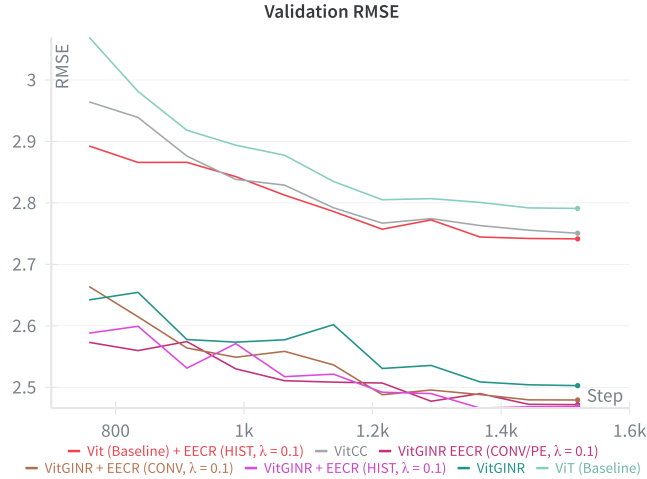


Figure 7: RMSE validation metric plot of the methods reported in Table 2.